Lecture 8

Agency and Moral Hazard

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Lotteries over Money

Let X be a continuous variable representing money

Describe a lottery by a CDF $F : \mathbb{R} \to [0,1]$; $F(x)$ is probability that payoff is less than or equal to x

If F has a derivative (PDF) f, then we can write

$$F(x) = \int_{-\infty}^{x} f(x)dx$$

Let $\mathcal{L}$ be the set of all CDFs over nonnegative sums (i.e. $\int_{-\infty}^{0} f(x)dx = 0$)

Assuming preferences satisfy expected utility axioms,[1] there is an assignment of utility values $u(\cdot)$ such that

$$U(F) = \int u(x)\, dF(x)$$

1. Add continuity and independence to completeness and transitivity. More details upon request.

Risk Aversion

A decision maker is **risk averse** if, for any lottery $F(\cdot) \in \mathcal{L}$, the degenerate lottery that yields $\int x \, dF(x)$ with certainty is at least as good as the lottery $F(\cdot)$ itself

A decision maker is risk averse if and only if

$$\int u(x)\, dF(x) \leq u\left(\int x \, dF(x)\right) \text{ for all } F(\cdot)$$

With expected utility, risk aversion is equivalent to concavity of $u(\cdot)$, by Jensen's Inequality

First-Order Stochastic Dominance

Want to describe a distribution F that yields unambiguously higher returns than a distribution G

Definition

The distribution $F(\cdot)$ **first-order stochastically dominates** $G(\cdot)$ if, for every *nondecreasing* function $u : \mathbb{R} \to \mathbb{R}$

$$\int u(x)\, dF(x) \geq \int u(x)\, dG(x)$$

Proposition

Proposition

Proposition 1. *The distribution* $F(\cdot) \succsim_{FOSD} G(\cdot)$ *if and only if* $F(x) \leq G(x)$ *for all* x

Consider uniform distribution with different supports: $[0,1]$ and $[0,2]$

Consider normal distributions with same variance but different means

For proof we'll restrict attention to lotteries L such that there exists a finite $\overline{x} > 0$ such that $L(\overline{x}) = 1$

Proof of Proposition: “Only If"

Show that $F(\cdot) \succsim_{FOSD} G(\cdot)$ implies $F(x) \leq G(x)$ for all x. Suppose not and derive contradiction

Denote $H(x) = F(x) - G(x)$, suppose $H(x') > 0$ for some x'. By $FOSD$, for any non-decreasing u

$$\int u(x)\, dH(x) \geq 0$$

Let $u(x) = 1$ for $x > x'$ and $u(x) = 0$ for $x \leq x'$

$$\int u(x)\, dH(x) = |F(x) - G(x)|_{x'}^{\infty} = (1 - 1) - H(x') = -H(x') < 0$$

So F did not FOSD G, since u is a non-decreasing function

Proof of Proposition: “If" (assume differentiable utility

Show that $F(x) \leq G(x)$ for all x implies $F(\cdot) \succsim_{FOSD} G(\cdot)$

Integrating by parts yields

$$\int u(x)\, dH(x) = [u(x)H(x)]_0^\infty - \int u'(x)H(x)\, dx$$

Since $H(0) = 0$ and $H(\overline{x}) = 0$, the first term is zero

Since $F(x) \leq G(x)$ for all x, $H(x) \leq 0$ for all x

Thus, since $u' \geq 0$, we have

$$\int u(x)\, dH(x) = -\int u'(x)H(x)\, dx \geq 0$$

This is the definition of FOSD

Second-Order Stochastic Dominance

FOSD is about a “greater than" relationship. Might also want a “less risky than" relationship

Definition

For any two distributions $F, G \in \mathcal{L}$, $F(\cdot)$ **second-order stochastically dominates** (or is less risky than) $G(\cdot)$ if for every nondecreasing, *concave* function $u : \mathbb{R}_+ \to \mathbb{R}$ we have

$$\int u(x)\, dF(x) \geq \int u(x)\, dG(x)$$

Useful Fact: SOSD is equivalent to $F(\cdot)$ crossing $G(\cdot)$ once (from below)

Hidden Action

Many situations of interest have hidden action

- Executive to bureaucrat
- Voter to politician
- Congress to executive

How does a principal give the right incentives for costly effort to an agent with hidden action?

A Basic Model

Two players: principal (P) and agent (A)

P offers a wage scheme to A, A chooses whether to accept it, and if so and how much effort to exert

Let $e \in \{e_L, e_H\}$ be the effort choice of the agent

Let π be the "profits" to P, drawn from $[\underline{\pi}, \overline{\pi}]$

π drawn from $f(\cdot|e)$, $f(\pi|e) > 0$ for all $e \in \{e_L, e_H\}$ and $\pi \in [\underline{\pi}, \overline{\pi}]$

Assume effort is productive in the following sense

$$F(\cdot|e_H) \succsim_{FOSD} F(\cdot|e_L)$$

Payoffs

Principal maximizes expectation of profits minus wages

Agent has expected utility function

$$u(w, e) = v(w) - c(e)$$

satisfying $v' > 0$, $v'' < 0$ and $c(e_H) > c(e_L)$

So principal rewards agent, agent's behavior affects principal but cannot be perfectly monitored

Complete Information Benchmark

Suppose principal offers a "contract" (fancy name for a reward scheme) and agent can accept or reject contract

If reject, agent gets reservation utility $\overline{u}$ and principal gets 0

A contract specifies effort and wage as a function of profits

Principal's optimal contract solves

$$\max_{e \in \{e_L, e_H\}, w(\cdot)} \int (\tilde{\pi} - w(\tilde{\pi})) f(\tilde{\pi}|e)\, d\tilde{\pi}$$

$$\text{s.t.} \int v(w(\tilde{\pi})) f(\tilde{\pi}|e)\, d\tilde{\pi} - c(e) \geq \overline{u}$$

Constrained Maximization: A Digression

$$\max_a f(a,\theta) \text{ s.t. } g(a) \geq \overline{g}$$

By Kuhn-Tucker (subject to technical conditions i.e. ‘constraint qualification’), we can set up the Lagrangian

$$L(a) = f(a,\theta) + \lambda(g(a) - \overline{g})$$

The Kuhn-Tucker conditions for a constrained optimum are:

$$\frac{\partial L}{\partial a}(a^*) = 0$$

$$\lambda(g(a^*) - \overline{g}) = 0$$

$$\lambda \geq 0$$

Full Information Optimal Contract

The contract involves a schedule of wages for each possible profit-effort pair

Given effort choice, want to minimize wages

How to do it?

- figure out desired effort, give zero wages for other efforts
- minimize wages subject to individual rationality constraint

Full Information Optimum (cont'd)

For a fixed e, the optimal $w(\cdot)$ minimizes compensation

$$\min_{w(\cdot)} \int w(\tilde{\pi}) f(\tilde{\pi}|e)\, d\tilde{\pi} \text{ s.t. } \int v(w(\tilde{\pi})) f(\tilde{\pi}|e)\, d\tilde{\pi} - c(e) \geq \overline{u}$$

Notice that the constraint must bind, otherwise could have lower wages while still having contract accepted

How do we solve for the function $w(\cdot)$? Point-wise

- Imagine a finite number of possible profits

An Example with Two Possible Profits

$$\min_{w_1,w_2} w_1 f(\pi_1|e) + w_2 f(\pi_2|e)$$

$$\text{s.t. } v(w_1)f(\pi_1|e) + v(w_2)f(\pi_2|e) - c(e) \geq \overline{u}$$

- The Lagrangian is

$$L = -\left[w_1 f(\pi_1|e) + w_2 f(\pi_2|e)\right] +$$
$$\lambda\left[v(w_1)f(\pi_1|e) + v(w_2)f(\pi_2|e) - c(e) - \overline{u}\right]$$

- From the Kuhn-Tucker conditions

$$-f(\pi_1|e) + \lambda v'(w_1)f(\pi_1|e) = 0 \iff \frac{1}{v'(w_1)} = \lambda$$
$$-f(\pi_2|e) + \lambda v'(w_2)f(\pi_2|e) = 0 \iff \frac{1}{v'(w_2)} = \lambda$$

- We will do the analogous thing for the continuous case (ignoring the measure-theoretic issues)

Full Information Optimum (cont'd)

$$L = -\int w(\tilde{\pi}) f(\tilde{\pi}|e)\, d\tilde{\pi} + \lambda \left(\int v(w(\tilde{\pi})) f(\tilde{\pi}|e)\, d\tilde{\pi} - c(e)\right.$$

- From Kuhn-Tucker at each π, $w(\pi)$ must satisfy

$$-f(\pi|e) + \lambda v'(w(\pi)) f(\pi|e) = 0 \iff \frac{1}{v'(w(\pi))} = \lambda$$

- **Optimal contract has wage (given effort) constant in profit**

Intuition for Risk Sharing

Contract specifies effort, so no incentive problem

Risk-neutral principal fully insures risk-averse agent

Principal must give the agent at least his reservation utility

The cheapest way to do this is to give him that payoff for sure

Full Information Optimal Contract, cont'd

So optimal contract offers a fixed wage, given effort

For a given effort level e^*, choose wage satisfying
$v(w(e^*)) - c(e^*) = \overline{u}$

$w(e^*) = v^{-1}(\overline{u} + c(e^*))$

What effort level to specify in contract?

$$\max_{e \in \{e_L, e_H\}} \int \tilde{\pi} f(\tilde{\pi}|e)\, d\tilde{\pi} - v^{-1}(\overline{u} + c(e))$$

Unobservable Effort

Optimal contract with observable effort does two things:

1. Specifies efficient effort choice by agent
2. Insures agent against all risk

If cannot contract on effort, these two may come into conflict

Only way to get agent to work hard is to relate pay to outcomes

Moral hazard will lead to inefficiency

Risk-Neutral Agent

Suppose $v(w) = w$, so agent risk neutral

- shouldn't matter that can't insure against risk

We can achieve same effort choice and expected utilities with no observability

[Like selling the enterprise to the worker: the principal takes a fixed amount and the agent gets everything else.]

Risk-Neutral Agent, cont'd

Let $w(\pi) = \pi - \alpha$ for a constant α

If agent accepts, chooses effort to maximize

$$\int w(\tilde{\pi}) f(\tilde{\pi}|e) d\tilde{\pi} - c(e) = \int \tilde{\pi} f(\tilde{\pi}|e) d\tilde{\pi} - \alpha - c(e)$$

The same e is chosen under observability!

What α will principal choose? Principal gets $\pi - w(\pi) = \alpha$, so choose the largest possible one

So, choose $\alpha = \alpha^*$ such that $\int \tilde{\pi} f(\tilde{\pi}|e^*) d\tilde{\pi} - \alpha^* - c(e^*) = \overline{u}$

Principal obtains $\alpha^* = \int \tilde{\pi} f(\tilde{\pi}|e) - c(e^*) - \overline{u}$. Agent obtains $\overline{u}$

Risk-Averse Agent

2 steps: Optimal wages given an effort. Then optimal effort

If principal wants to implement e must satisfy two constraints

1. **Individual Rationality:** Must accept contract
2. **Incentive Compatibility:** Must prefer e to alternative

$$\min_{w(\cdot)} \int w(\tilde{\pi}) f(\tilde{\pi}|e)\, d\tilde{\pi} \text{ s.t}$$

$$\int v(w(\tilde{\pi})) f(\tilde{\pi}|e)\, d\tilde{\pi} - c(e) \geq \overline{u}$$

$$e \in \arg\max_{\hat{e}} \left\{ \int v(w(\tilde{\pi})) f(\tilde{\pi}|\hat{e})\, d\tilde{\pi} - c(\hat{e}) \right\}$$

Implementing Low Effort

fixed wage $v^{-1}(\overline{u} + c(e_L))$

Incentive Compatibility: Agent prefers e_L to e_H, since wage is unaffected by outcome or effort. Therefore no need to worry about IC in the Lagrangean.

Individual Rationality: Agent earns exactly $\overline{u}$

Optimality for Principal: Any lower wage would fail IR

Implementing High Effort

Rewrite IC constraint as

$$\int v(w(\tilde{\pi}))f(\tilde{\pi}|e_H)\,d\tilde{\pi} - c(e_H) \geq \int v(w(\tilde{\pi}))f(\tilde{\pi}|e_L)\,d\tilde{\pi} -$$

The Lagrangian is then

$$L = -\int w(\tilde{\pi})f(\tilde{\pi}|e_H)d\tilde{\pi}$$
$$+\lambda_1\left[\int v(w(\tilde{\pi}))f(\tilde{\pi}|e_H)d\tilde{\pi} - c(e_H) - \overline{u}\right]$$
$$+\lambda_2\left[\int v(w(\tilde{\pi}))[f(\tilde{\pi}|e_H) - f(\tilde{\pi}|e_L)]d\tilde{\pi} - c(e_H) + c(e_L)\right]$$

Implementing High Effort

Rewrite IC constraint as

$$\int v(w(\tilde{\pi}))f(\tilde{\pi}|e_H)\,d\tilde{\pi} - c(e_H) \geq \int v(w(\tilde{\pi}))f(\tilde{\pi}|e_L)\,d\tilde{\pi} -$$

Using point-wise minimization, from Kuhn-Tucker, at every π we need

$$-f(\pi|e_H) + \lambda_1 v'(w(\pi))f(\pi|e_H)$$
$$+\lambda_2\left(f(\pi|e_H) - f(\pi|e_L)\right)v'(w(\pi)) = 0$$

or

$$\frac{1}{v'(w(\pi))} = \lambda_1 + \lambda_2\left(1 - \frac{f(\pi|e_L)}{f(\pi|e_H)}\right)$$

Both Constraints Bind

$$\frac{1}{v'(w(\pi))} = \lambda_1 + \lambda_2 \left(1 - \frac{f(\pi|e_L)}{f(\pi|e_H)}\right)$$

Suppose $\lambda_1 = 0$

- Since $F(\cdot|e_H) \succsim_{FOSD} F(\cdot|e_L)$, there is an open set Π such that $f(\pi|e_L) > f(\pi|e_H)$ for $\pi \in \Pi$
- But then if $\lambda_1 = 0$ for $\pi \in \Pi$, $v' \leq 0$ which is impossible

Suppose $\lambda_2 = 0$

- Optimal schedule gives fixed wages for all π
- Agent chooses e_L

Wage Schedule and Incentives for Effort

$$\frac{1}{v'(w(\pi))} = \lambda_1 + \lambda_2 \left(1 - \frac{f(\pi|e_L)}{f(\pi|e_H)}\right)$$

Consider $\hat{w}$ such that $1/v'(\hat{w}) = \lambda_1$

$$w(\pi) > \hat{w} \quad \text{if} \quad \frac{f(\pi|e_L)}{f(\pi|e_H)} < 1$$

$$w(\pi) < \hat{w} \quad \text{if} \quad \frac{f(\pi|e_L)}{f(\pi|e_H)} > 1$$

Pay more than $\hat{w}$ for outcomes that are relatively more likely to occur under e_H than under e_L (likelihood ratio less than 1)

Even though can anticipate effort choice under contract

Shape of Wage Schedule

Rewarding good news about e_H creates incentives for e_H

- Reward outcomes that are sensitive to effort

Does not imply compensation increasing in profits!

Compensation is increasing in profit if likelihood ratio $\frac{f(\pi|e_L)}{f(\pi|e_H)}$ is decreasing (Monotone Likelihood Ratio Property)

MLRP stronger than FOSD

Even without MLRP, might get increasing wage schedules due to agent's ability to engage in fraud (destroy product if get an unfavorable production level)

FYI

Families of distributions that satisfy MLRP:

- Normals with different μ
- Binomial with different p
- Exponential with different λ
- Poisson with different λ

What Level of Effort is Induced?

If want e_L give same incentives as in complete information case

If want e_H must compensate for risk with higher expected wage payment

Optimal contract implements e_H less frequently (there are situations in which e_H is implemented under observable effort but not under unobservable effort)

Continuous Actions

Consider the standard moral hazard problem with risk neutral principal and risk averse agent

Suppose now that the agent can choose an action $e \in \mathbb{R}_+$

How do we approach this problem?

The Principal's Problem

Same as before

$$\min_{w(\cdot)} \int w(\tilde{\pi}) f(\tilde{\pi}|e)\, d\tilde{\pi} \text{ s.t}$$

$$\int v(w(\tilde{\pi})) f(\tilde{\pi}|e)\, d\tilde{\pi} - c(e) \geq \overline{u}$$

$$e \in \arg\max_{e' \in \mathbb{R}_+} \int v(w(\tilde{\pi})) f(\tilde{\pi}|e')\, d\tilde{\pi} - c(e')$$

Incentivizing an Arbitrary e

Suppose want agent to choose e^*

e^* must satisfy FOC of IC:

$$\int v(w(\tilde{\pi}))f_e(\tilde{\pi}|e^*)d\tilde{\pi} - c'(e^*) = 0$$

Notice, $w(\cdot)$ can be poorly behaved, so we don't know this identifies a maximum

If it does we can set up our Lagrangian, substituting the FOC for the IC, and optimize pointwise

Optimal Contract with FOA

$$\frac{1}{v'(w(\pi))} = \lambda_1 + \lambda_2 \left(1 - \frac{f_e(\pi|e^*)}{f(\pi|e^*)}\right)$$

- This is just as before (as long as FOA is valid)
- $\frac{f_e}{f}$ decreasing is implied by MLRP
- There are well known (but somewhat restrictive) conditions under which the FOA is valid

Moral Hazard with Extra Signal

Suppose principal observes π and also some additional signal y

y doesn't enter directly into the principal's welfare

Should the principal offer wages $w(\pi, y)$ which depend on y?

Only if y provides *extra* information about effort choice

Sufficient Statistic Result

Suppose the principal wants to incentivize high effort, wages satisfy

$$\frac{1}{v'(w(\pi, y))} = \lambda_1 + \lambda_2 \left(1 - \frac{f(\pi, y|e_L)}{f(\pi, y|e_H)}\right)$$

Wages are the same as without y if and only if (ignoring some measure theoretic concerns) for all y

$$\frac{f(\pi, y|e_L)}{f(\pi, y|e_H)} = \frac{f(\pi|e_L)}{f(\pi|e_H)}$$

The above condition is equivalent to $f(e|\pi, y) = f(e|\pi)$ for all values of y

In words: π is sufficient statistic for (π, y) with respect to e

Say what?

What is a Sufficient Statistic?

Consider the maximum likelihood problem (i.e., trying to figure out the e chosen from the π observed)

Your best guess satisfies

$$\max_e \ln f(\pi|e)$$

The FOC is:

$$\frac{f_e(\pi|e)}{f(\pi|e)} = 0$$

π is a sufficient statistic for (π, y) with respect to e, if and only if the maximum likelihood estimator $\max_e \ln f(\pi, y|e)$ is the same as the maximum likelihood estimator above (ignoring a bit of measure theoretic concerns)